ATS

ALL TECHNICAL SOLUTIONS

MAINTENANCE GUIDE

Drive Replacement Procedure

TrueNAS SCALE Dragonfish 24.04 | Pool: CAVO-POOL

CAVO Design — Server: 10.1.4.51

DRIVE DETAILS

Server:	**CAVO-FN002 (10.1.4.51)**
Pool:	**CAVO-POOL**
Replacement:	**4x WD Red Plus 4TB (new drives staged on-site)**
OS:	**TrueNAS SCALE Dragonfish 24.04**
ATS Contact:	**Marty Bauer | 215-469-1287**

REPLACE ONE DRIVE AT A TIME -- WAIT FOR RESILVER BEFORE THE NEXT

QUICK REFERENCE -- At a Glance

Server IP	**http://10.1.4.51**
Pool name	CAVO-POOL
New drives	4x WD Red Plus 4TB (staged and ready)
Drive at risk	1 pending sector detected on /dev/sdf -- replace this one FIRST
ATS Contact	**Marty Bauer | 215-469-1287 -- stay on Zoom throughout**
Rule #1	Replace ONE drive at a time. Wait for resilver to complete before the next.
Rule #2	NEVER power off the server while resilvering.
Rule #3	Anything looks wrong -- STOP and call Marty.

PROCEDURE OVERVIEW

1	Log in to TrueNAS web UI at http://10.1.4.51
2	Check pool status -- note which drive(s) show errors
3	Identify the drive physically using the LED blink method
4	Offline the drive in TrueNAS before touching hardware
5	Physically remove old drive and insert new drive
6	Replace drive in TrueNAS -- resilver starts automatically
7	Monitor resilver (3-8 hours) -- pool must show HEALTHY before continuing
8	Repeat Steps 3-7 for each remaining drive

BEFORE YOU BEGIN -- Checklist

Complete every item before touching any drives. Marty should already be on Zoom or phone.

- [ ] All 4 new WD Red Plus 4TB drives are physically on-site.
- [ ] You can open a browser on a computer connected to the CAVO network.
- [ ] Marty is reachable at 215-469-1287 and on Zoom if possible.
- [ ] No urgent work is being done on the server right now.
- [ ] You can physically access the drive bays on CAVO-FN002.
- [ ] You have a flashlight available.

> **Hot-swap -- the server stays ON during the entire procedure**
> Removing a drive while powered is intentional and safe on this hardware.
> The server and all files remain accessible throughout the process.
> Handle drives carefully: hold by the sides, avoid touching the circuit board.

1 Log in to TrueNAS Web UI

On any computer on the CAVO network, open a web browser and navigate to:

```
http://10.1.4.51
```

Log in with the TrueNAS root credentials. If you don't have the password, ask Marty.

2 Check Pool Status

Click **Storage** in the left-hand menu. Find the **CAVO-POOL** tile. The status badge tells you the current health:

Status	Meaning	Action
HEALTHY	Pool is fine -- no data at risk	Proceed normally
DEGRADED	A drive has already failed	Proceed -- replace the failed drive first
FAULTED	Critical -- multiple failures	STOP. Do not touch. Call Marty.

3 Identify the Drive to Replace -- BLINK THE LED

This is the most important step. You must identify the exact physical drive bay before removing anything. Two methods:

Method A -- TrueNAS Identify Button (try first)

1. In TrueNAS, click **Storage**, then click **Disks** (link near top of page).
2. Find the drive that shows errors or was flagged as failing.
3. Click the **three-dot menu (...)** on that drive's row.
4. Click **Identify** -- the bay LED on the physical server will flash for 15 seconds.
5. Watch the drive bay lights on the server -- the flashing bay is your target.
6. Note the bay position or place a small piece of tape on it before the LED stops.

Method B -- Shell command (if Identify button is not available or nothing blinks)

```
dd if=/dev/sdf of=/dev/null bs=1M status=none &
```

The activity LED for **/dev/sdf** starts blinking rapidly. That is your drive.
Once identified, type **kill %1** in the shell to stop it (or press Ctrl+C).
To confirm by serial number: smartctl -i /dev/sdf **|** grep Serial

4 Offline the Drive in TrueNAS (Before Touching Hardware)

Tell TrueNAS to stop reading/writing to this drive before you physically remove it. This prevents errors.

Storage	Click **Storage** in the left menu.
Gear	Click the **gear icon** on the CAVO-POOL tile (top-right of the tile).
Status	Click **Status** -- you will see a tree listing all drives in the pool.
Find	Locate the failing drive in the tree (matches the /dev/sdX name from Step 3).
Menu	Click the **three-dot menu (...)** next to that drive.
Offline	Click **Offline**. Confirm when prompted.
Confirm	Drive status changes to **OFFLINE**. Pool may show DEGRADED -- this is expected.

5 Physical Drive Swap

> **The server stays ON**
> Hot-swap = removing a drive while powered is safe and intentional on this hardware.
> Work slowly. Never force the tray. Handle drives by the edges only.

1. Go to the server and locate the drive bay you identified in Step 3.

2. Press or slide the **release latch** on the drive tray (small lever on the front face of the tray).

3. Slide the tray **slowly and straight out**. Set it on a flat surface.

4. Unscrew or unclip the old drive from the tray (typically 4 screws on the sides or bottom).

5. Remove the old drive. Set it aside clearly labeled -- do NOT discard it.

6. Place the **new WD Red Plus 4TB** into the tray, aligning the screw holes.

7. Secure the new drive in the tray with the same screws or clips.

8. Slide the tray firmly back into the bay until you feel it **click into place**.

9. The bay light should turn on -- TrueNAS is detecting the new drive.

6 Replace Drive in TrueNAS -- Start Resilver

Now assign the new drive to the pool slot and start the rebuild (called **resilvering**).

1. In TrueNAS, go to **Storage -> CAVO-POOL gear icon -> Status**.

2. Find the drive slot showing **OFFLINE** (the one you just swapped).

3. Click the **three-dot menu (...)** next to it.

4. Click **Replace**.

5. In the dialog, click the **Member Disk** dropdown.

6. Select the new drive (it appears as an unformatted/blank disk).

7. Click **Replace Drive** to confirm.

8. Pool status changes to DEGRADED with a progress bar -- **NORMAL. Resilver is running.**

7 Monitor Resilver Progress

TrueNAS copies all pool data to the new drive. A 4TB drive typically takes **3 to 8 hours**.

How to check	Storage -- CAVO-POOL tile shows progress bar + percentage
During resilver	Pool shows DEGRADED -- this is NORMAL. Server is fully operational.
When finished	Pool returns to HEALTHY (green checkmark). Then and ONLY then proceed to the next drive.

If stalled	Progress doesn't change for 30+ minutes -- call Marty.

Resilver is running in the background -- you can leave and come back

Files remain accessible. Users can work normally.

Check back every hour or two. Marty will confirm when to proceed to the next drive.

8 Repeat for Remaining Drives

Only after pool shows HEALTHY -- repeat Steps 3 through 7 for each drive. Replacing all 4 drives one at a time.

Round	Drive to Replace	Done?	Notes
1 -- FIRST	Failing drive (/dev/sdf)	[ ]	PRIORITY -- has pending sector
2	Next oldest drive	[ ]	Marty will advise which
3	Next drive	[ ]	
4 -- LAST	Last drive	[ ]	

EMERGENCY REFERENCE -- Stop and Call Marty

STOP -- CALL MARTY IMMEDIATELY: 215-469-1287

TrueNAS shows **FAULTED** pool status at any point.

Two or more drives show errors or fail simultaneously.

New drive inserted but TrueNAS does not detect it after 2 minutes.

Resilver progress shows 0% and does not move for over 30 minutes.

Any error message appears that you don't understand.

Drive tray will not slide back in or does not click into place.

Server shuts off or loses power unexpectedly.

Any doubt at all -- stop and call. There are no dumb questions.

TrueNAS Shell Commands -- Marty's Reference (open via Shell icon in top menu)

COMPLETION CHECKLIST

[ ] All 4 drives replaced, one at a time, with resilver completing between each.

[ ] Final CAVO-POOL status is HEALTHY (green).

[ ] Old drives set aside and labeled -- not discarded (Marty may need them).

[ ] Short SMART test run on all 4 new drives (Storage -> Disks -> Run Manual Test -> Short).

[ ] Marty confirmed via phone/Zoom that job is complete.

[ ] Completion date and time noted for billing.

After the job -- Marty's next steps

Run a full overnight SMART long test on all new drives.

Verify Immich, replication, and all apps are still running.

Set up the ATS Network Monitor portal access for CAVO.

Provide Marc Caccavo the Frigate camera feed link and portal QR code.

Configure UPS graceful shutdown (see next page).

UPS CONFIGURATION -- Graceful Shutdown Setup

A power outage without a configured UPS can corrupt the ZFS pool mid-write. TrueNAS SCALE has NUT (Network UPS Tools) built in. Configure it once after the drives are installed and the pool is HEALTHY.

> **STOP -- CALL MARTY IMMEDIATELY: 215-469-1287**
>
> **Lesson learned:** Power outage during or before drive installation can interrupt a resilver and risk data loss. Complete UPS setup before any future maintenance window.

STEP-BY-STEP -- TrueNAS SCALE UPS Setup

1 Connect UPS to server via USB

Plug the USB monitoring cable from the UPS into a USB port on CAVO-FN002. The UPS must have a USB HID port (most APC and CyberPower units have one).

2 Open TrueNAS UPS settings

In TrueNAS UI: System -> Services -> UPS -> Configure (pencil icon).

3 Set UPS Mode

Mode: **Master** | Identifier: **cavo-ups** | Driver: leave as Auto-detect.

4 Set shutdown behavior

Shutdown Mode: **UPS reaches low battery**
Shutdown Timer: **30 seconds**
Power Off UPS after shutdown: **CHECKED** -- prevents the server from restarting automatically on power restore (important if a resilver is in progress).

5 Enable the service

Toggle UPS service **ON**. TrueNAS will detect the connected UPS and show battery level and status immediately.

6 Verify UPS is detected

System -> Services -> UPS should show **Status: Online** with a battery percentage. Also visible under Reporting -> UPS.

7 Test shutdown (do during a maintenance window)

Unplug UPS from wall. Watch battery % drop in TrueNAS. TrueNAS should initiate a graceful shutdown before the battery reaches critical. Reconnect power -- server should restart cleanly with pool HEALTHY.

WireGuard note -- power outage behavior